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FOSS4G EU - JULY 2026

EOPF

FOSS4GEU26

EXPLORER

Web Visualization Techniques and Resources for the GeoZarr Specification

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EOPF EXPLORER

FOSS4G EU - JULY 2026

1 Get set up

What you'll learn today

2 Introduction to EOPF

Why cloud-native

3 What is GeoZarr

Problems it solves

4 Data Access

EOPF Explorer STAC API

1 Get set up

What you'll learn today

WHAT YOU'LL LEARN TODAY

Get Set up

- https://github.com/EOPF-Explorer/foss4g-eu-26-workshop

What you'll learn today

- Read cloud-native GeoZarr straight from S3
- Style and combine bands in the browser
- Render server-side tiles and band-math (NDVI, SWIR) with TiTiler
- Build a scrollytelling story
- Build a STAC-driven dashboard
- Visualize Interactive components in Jupyter notebooks

2 Introduction to EOPF

Why cloud-native

WHAT IS EOPF?

Earth Observation Processing Framework

- ESA's initiative for modernised Copernicus data
- A new approach to processing and distributing Sentinel data

Standardised Format Across Missions

- Single meta-model applicable to all Sentinel-1, -2, and -3 (land) missions
- Consistent structure regardless of which satellite captured the data

EOPF PRODUCT

- MEASUREMENTS
- QUALITY
- CONDITIONS
- ATTRIBUTES

CLOUD-NATIVE: THE BENEFITS

Traditional Workflow

- Download entire datasets (often 500MB to several GB per scene)
- Store locally before analysis
- Limited by storage capacity and bandwidth

Cloud-Native Workflow

- Dynamic rendering without downloads
- Access only the data you need, when you need it
- Work directly with data in the cloud and scale up your analysis

WHAT IS ZARR?

A Cloud-Optimised Data Format

- Multi-dimensional array storage format
- Designed specifically for cloud environments
- Open specification with growing ecosystem

Key Technical Features

Chunking: Data divided into manageable pieces

Partial retrieval: Access only what you need

Parallel access: Multiple chunks can be read simultaneously

Compression: Efficient storage without sacrificing access speed

RESOURCES FOR THE TRANSITION TO EOPF

TOOLKIT

EOPF

LEARN THE BASICS

SENTINEL ZARR SAMPLES

EOPF

DISCOVER THE DATA

VISUALISE: TELL A STORY

EXPLORER

EOPF

EOPF EXPLORER

3 What is GeoZarr

What does it solve? and how?

GEOZARR SPECIFICATION

🌍 GeoZarr Specification

OGC Geospatial Zarr Standard

📐 Multiscales

Encoding pyramid information for multiresolution data

🗺️ Projection

CRS and datum encoding for geospatial data

📍 Spatial

Array indices to spatial coordinates relationship

Zarr v3 Core

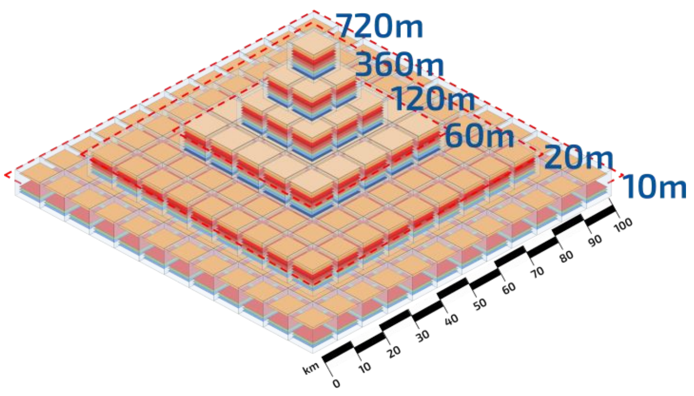
MULTISCALES ATTRIBUTE EXTENSION

The problem: one full-resolution band is huge

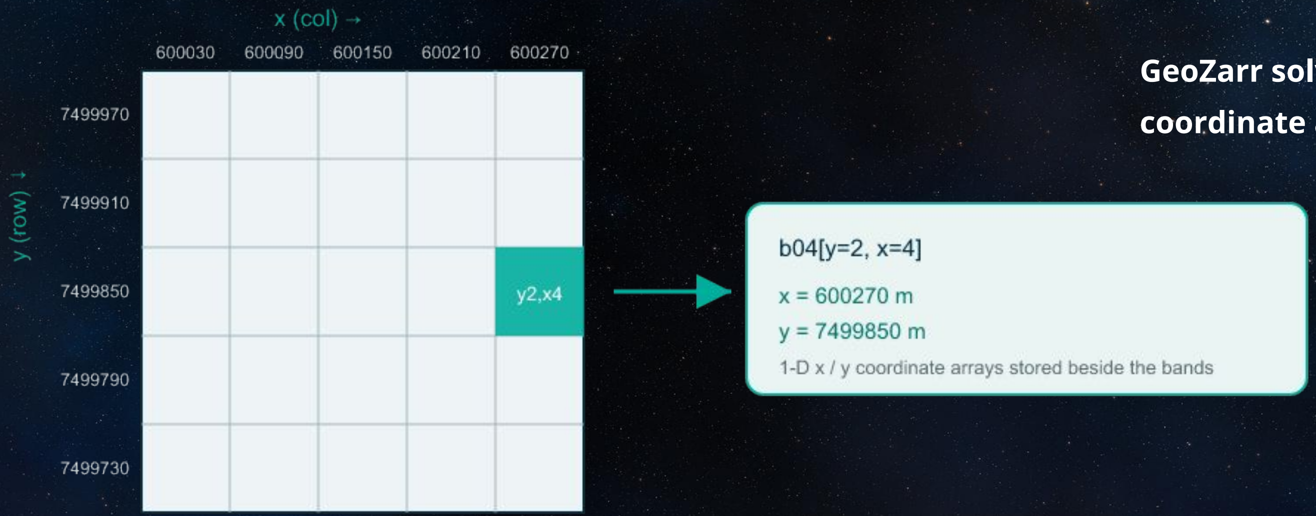
- One 10 m band is 10980×10980 ≈ 120 MP ≈ 480 MB

GeoZarr solves it with an overview pyramid

- This scene: six levels
- r10m → r20m → r60m → r120m → r360m → r720m
- r60m is 1830×1830; r720m just 152×152



SPATIAL CONVENTION



The problem: which axis is which, and where's this pixel?

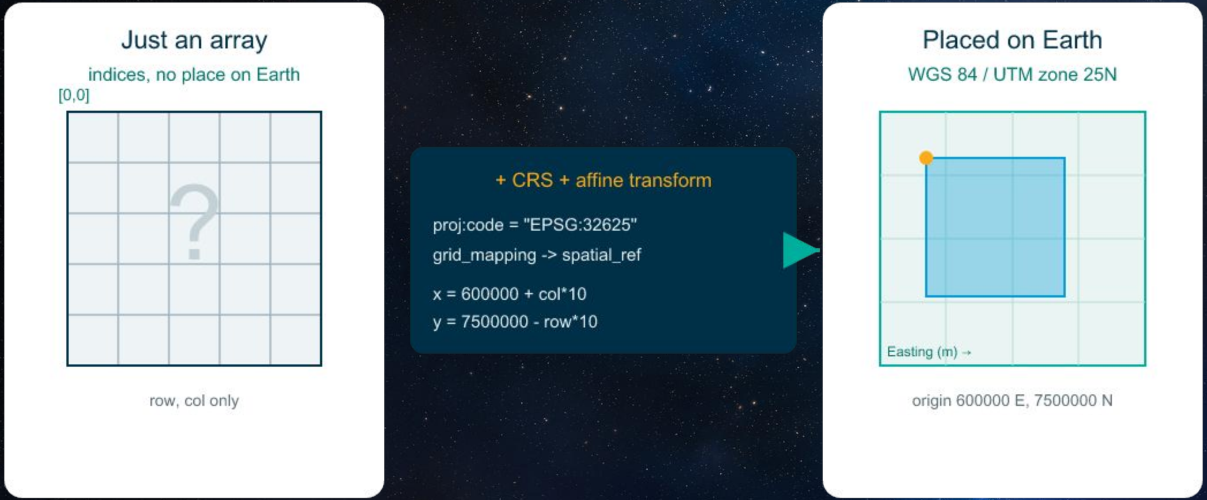
GeoZarr solves it via named axes and coordinate arrays.

GEOSPATIAL PROJECTION ATTRIBUTE EXTENSION

The problem: How to position on an Earth map?

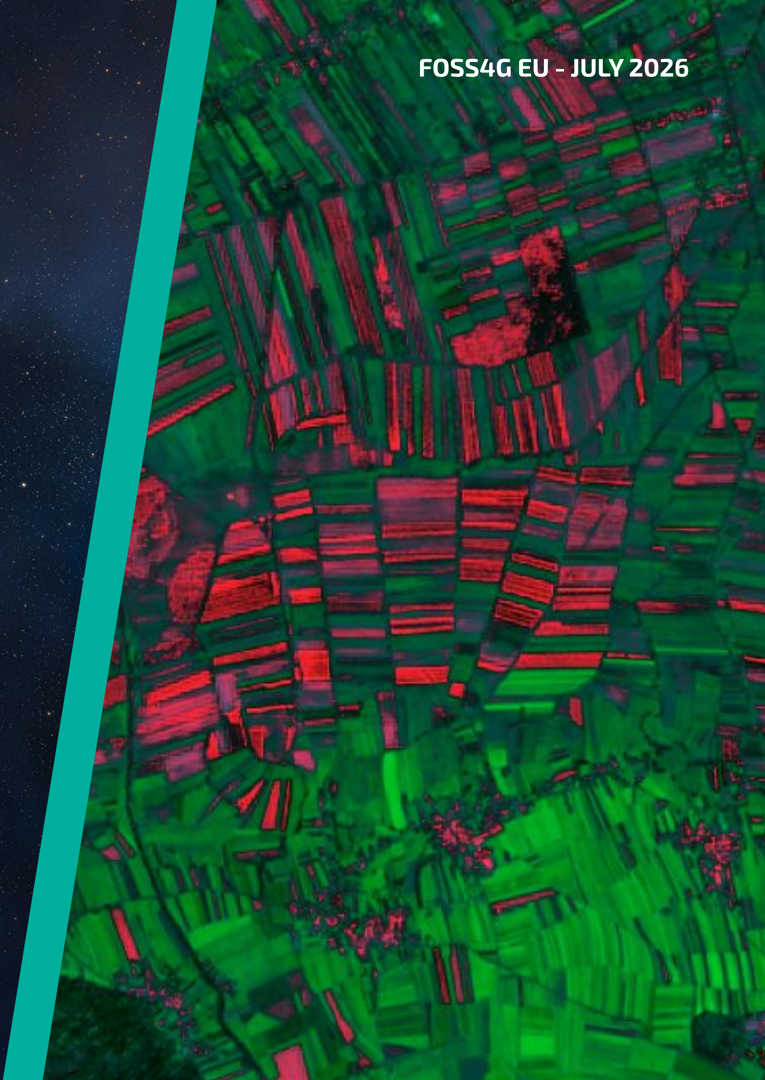
- Indices are row, col
- You can't overlay, reproject, or measure

GeoZarr solves it by CRS and an affine transform.

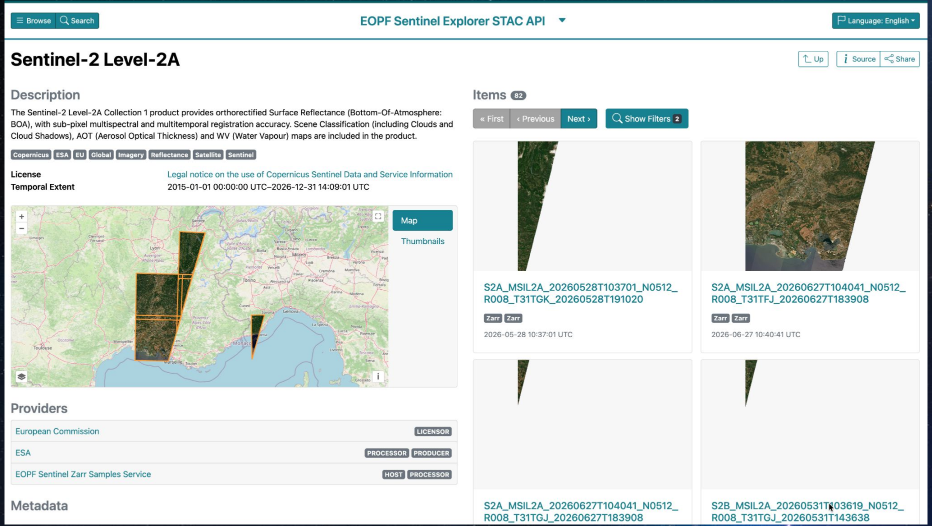


4 Data Access

EOPF Explorer STAC API



FIND A GEOZARR IN THE EOPF EXPLORER STAC



Browse the catalogue

- STAC browser: api.explorer.eopf.copernicus.eu/browser
- API: api.explorer.eopf.copernicus.eu/stac
- Collection: sentinel-2-l2a

Filter down to exactly one scene.

- bbox + datetime + eo:cloud_cover → the newest matching item
- One item = one acquisition = one .zarr store

INSIDE THE ITEM - ASSETS

STAC item › assets

```
"assets": {
  "reflectance": {
    "href": "…/measurements/reflectance"
    "type": "application/vnd.zarr; version=3; profile=multiscales"
    "roles": ["data", "reflectance"]
  },
  "SCL_20m": "…/conditions/mask/l2a_classification/r20m/scl"
  "AOT_10m": "…/quality/atmosphere/r10m/aot"
  "WVP_10m": "…/quality/atmosphere/r10m/wvp"
}
```

The GeoZarr reflectance group

media type carries the multiscales profile

href goes straight into eox-map

Other groups in the same store

SCL — scene classification

AOT · WVP — atmosphere

INSIDE THE ITEM - LINKS

STAC item › links

"links": [
{ "rel": "store", "href": "https://s3.…/S2A_…_T28WDD_….zarr" }
{ "rel": "xyz", "href": "…/tiles/WebMercatorQuad/{z}/{x}/{y}.png"
?variables=/measurements/reflectance:b04 …b03 …b02 }
{ "rel": "tilejson", "href": "…/WebMercatorQuad/tilejson.json" }
{ "rel": "viewer", "href": "…/items/…/viewer" }
{ "rel": "via", "href": "explorer.eopf.copernicus.eu/…/items/…" }
]

Root .zarr store

open it directly with zarrita.js

TiTiler XYZ tiles

{z}/{x}/{y}.png, bands pre-combined

drop into any web map — no client GPU